Study Guide for Exam #2

MATH 240 — Introduction to Statistics, Fall 2026

General Information

The second exam will be held in-class on Friday, November 13th. Keep in mind the following things:

- You will have 50 minutes to complete the exam, unless you have an accommodation.
- This is a **closed-book** exam, with the exception of two 8.5x11" sheets of paper of notes. You can use both sides of the sheets to write down anything.
- You will not need to write any R code on this exam. However, you must be able to interpret R output (e.g. plots, metrics, probabilities, etc.).
- Cell-phones, laptops, calculators, and any other electronic devices are not permitted. No interaction with anyone is allowed.

Expect the exam to be divided up into three parts:

- Part I: Multiple choice
- Parts II and III: Short answers based on a situation/dataset

There will be about 5-10 multiple choice questions, followed by 3-4 questions for each of the remaining parts. All of the topics below are fair game for these questions.

Note that some topics from before the first exam may show up implicitly from the analysis we are doing. For instance, the sampling distributions we've seen in this unit might involve some visual analysis (e.g. center, shape, and spread), which we've discussed during EDA.

Topics

- Ch. 8: Multiple Linear Regression
  - Finding least-squares estimates (and their interpretations)
  - Diagnosing a model fit
  - Interpreting least-squares estimates and finding significance

  - Performing forward and backward selection using R^2_{adj}
  - Predicting $\hat{y}$ using the least squares line

- Ch. 9: Logistic Regression and Probability
  - Writing the logistic regression model and finding probabilities using the model
  - Finding the odds ratio for a covariate
  - Classify new observations using the estimate logistic regression model
  - Finding probabilities for events from a sample space

- Ch. 11: Hypothesis Testing (with Randomization)
  - Define and perform the five steps of hypothesis testing
  - Using randomization testing for an experiment

- Ch. 12: Confidence Intervals (with Bootstrapping)
  - Define a confidence interval and how they are constructed
  - Using bootstrapping to generate confidence intervals

- Ch. 13: Inference with Mathematical Models
  - Using the normal distribution to calculate probabilities, such as z-scores and the empirical rule
  - Connect the use of a normal distribution with a sampling distribution

- Parts of Ch. 14: Decision Errors
  - Connecting the relationship between discernibility and power to sample characteristics, such as sample size and statistics

Practice Problems

1. Ch. 8: Exercises 4, 7, 9, 11, 13, 15
2. Ch. 9: Exercises 5, 7
3. Ch. 11: Exercises 2, 4, 5, 7
4. Ch. 12: Exercises 3, 4, 5, 7
5. Ch. 13: Exercises 1, 3, 5, 9, 15
6. Ch. 14: Exercise 1